

Output

1. Native Simulation

The application was successfully built and executed using Zephyr's native_sim target. The terminal confirms that Zephyr OS v4.2.0 boots successfully.

```
1-- Zephyr version: 4.2.0 (/home/sanathkumar/deps/zephyr), build: v4.2.0
1[94/96] Linking C executable zephyr/ze...onal byproducts on target: zephyr_pre0
Generating files from /home/sanathkumar/zephyr-course/build/zephyr/zephyr.elf fo
r board: native_sim
[96/96] Running utility command for native_runner_executable
((.venv) ) sanathkumar@luffy:~/zephyr-course$ west flash
-- west flash: rebuilding
[2/2] Running utility command for native_runner_executable
PR -- west flash: using runner native
se *** Booting Zephyr OS build v4.2.0 ***
```

2. STM32F446RE Hardware

The LED blink application was successfully flashed to the NUCLEO-F446RE development board using west flash. The STM32F446xx device was detected correctly, the HEX file was programmed successfully, and the application started successfully.

```
Board      : NUCLEO-F446RE
Voltage    : 3.25V
SWD freq   : 4000 KHz
Connect mode: Under Reset
Reset mode  : Hardware reset
Device ID   : 0x421
Revision ID : Rev A
Device name : STM32F446xx
NVM size    : 512 KBytes
Device type : MCU
Device CPU   : Cortex-M4
BL Version  : 0x90
```

Opening and parsing file: zephyr.hex

Memory Programming ...

```
File       : zephyr.hex
Size       : 26.99 KB
Address    : 0x08000000
```

Erasing memory corresponding to segment 0:

Erasing internal memory sectors [0 1]

Download in Progress:

[=====] 100%

File download complete

Time elapsed during download operation: 00:00:00.870

RUNNING Program ...

```
Address:    : 0x8000000
```

Application is running, Please Hold on...

Start operation achieved successfully

((.venv)) sanathkumar@luffy:~/zephyr-course\$ █